Machine Learning-Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

Abstract

Real estate buyers differ in demographics, acquisition purpose, financing choices and geographic origins. This project implements an exploratory buyer-segmentation workflow and Streamlit dashboard using client and property records. Categorical variables are one-hot encoded, age is standardized, and categorical feature blocks are weighted to reduce the effect of high-cardinality fields. The recorded weighted K-Means run uses two clusters and excludes satisfaction_score from the final clustering representation because an earlier experiment showed that satisfaction dominated separation. Cluster 0 contains 969 buyers (48.45%) and cluster 1 contains 1,031 buyers (51.55%). The silhouette score is 0.3012, Davies–Bouldin index 1.2913 and Calinski–Harabasz score 1168.6032. The strongest descriptive distinction is age: mean ages are 38.80 and 68.48 years. The dashboard provides cluster distribution, acquisition-purpose and financing patterns, regional analysis and segment summaries. Results are exploratory; the data contain no ground-truth segment labels, income, net worth or investment-return outcomes.

Keywords: Buyer segmentation; real estate analytics; K-Means; feature encoding; customer profiling; Streamlit; unsupervised learning

1. Introduction

Real estate organizations serve buyers purchasing for personal use and investment. These groups may differ in demographics, financing, geographic origin and referral channel. A consistent analytical framework can help inspect these patterns and develop focused engagement hypotheses. The project brief identifies a need for data-driven understanding of buyer types, acquisition motivations, geographic differences and financing patterns. This study implements an unsupervised workflow and interactive dashboard based on the available schemas. The objective is exploratory intelligence, not a validated investment recommendation engine.

2. Objectives

The objectives are to clean and validate buyer records; transform mixed categorical and numeric fields; evaluate clustering with internal metrics; interpret groups through demographics, acquisition purpose, financing and geography; and deliver an interactive Streamlit dashboard with filters and downloadable summaries.

3. Dataset Description

The client table contains 2,000 rows and 12 fields: client_id, client_type, first_name, last_name, date_of_birth, gender, country, region, acquisition_purpose, satisfaction_score, loan_applied and referral_channel. The property table contains 10,000 rows and 9 fields: listing_id, tower_number, transaction_date, unit_category, unit_number, floor_area_sqft, sale_price, listing_status and client_ref. The client table has 2,000 unique IDs. Property status counts are 7,305 Sold and 2,695 Available; 2,695 property records have missing client_ref.

Equal counts do not prove one-to-one correspondence. Property analyses should use only reliably linked records. Currency is not assumed because the schema does not define it.

4. Cleaning and EDA

Categorical text is trimmed and empty values are treated as missing. Client rows are deduplicated by client_id. Age is derived from date_of_birth using 1 January 2025 as a fixed reference date. Implausible ages are treated as missing. Numeric values are converted safely and missing values are imputed in the preprocessing pipeline. Exploratory checks cover uniqueness, missingness, category counts, acquisition purpose, loan status and region. Recorded sale_price statistics are mean 344,374.68, median 330,680.845, standard deviation 131,563.10, minimum 97,402.80 and maximum 736,652.27. These are descriptive statistics, not currency-specific valuations.

5. Methodology

The clustering input uses client_type, gender, country, region, acquisition_purpose, loan_applied, referral_channel and age. Identifiers, names and raw date of birth are excluded. Categorical variables are one-hot encoded; age is median-imputed and standardized. Satisfaction was excluded from the final clustering representation after an earlier experiment showed it dominated between-cluster variance. Each categorical block is scaled by the square root of its encoded dimensionality. The saved configuration uses K-Means with K=2 and random_state=42. The project brief recommends hierarchical clustering as a complementary approach; comparative hierarchical results should only be reported after reproducible evaluation on the same processed dataset.

6. Evaluation

The recorded weighted K-Means solution has silhouette 0.3012, Davies–Bouldin 1.2913 and Calinski–Harabasz 1168.6032. These are internal clustering measures, not classification accuracy. Silhouette summarizes cohesion and separation; lower Davies–Bouldin is generally preferred; higher Calinski–Harabasz is generally preferred under comparable conditions. The metrics do not show that segments predict revenue, conversion or investment returns. Cluster IDs are arbitrary and can change when refitting.

7. Results

Cluster 0 contains 969 buyers (48.45%) and cluster 1 contains 1,031 buyers (51.55%). Mean ages are 38.80 and 68.48 years, with medians 39 and 68. Mean satisfaction is approximately 3.02 and 3.03, but satisfaction is not used to form the final clusters. Personal-use acquisition shares are 66.67% and 71.68%; investment-purpose shares are 33.33% and 28.32%. Loan-applied shares are 39.01% and 34.72%. Gender distributions are similar and both groups are predominantly individual clients. These are descriptive associations, not evidence that one group is more valuable. Labels such as “Luxury Investor” and “High-Net-Worth Investor” are unsupported because income, net worth and luxury status are not fields in the dataset.

8. Dashboard

The Streamlit dashboard includes Buyer Segmentation Overview, Investor Behavior Dashboard, Geographic Buyer Analysis, Segment Insights Panel, Customer Explorer and Methodology & Validation. It supports filters for country, region, acquisition purpose and client type. Property charts appear when property records can be

linked through client_ref. The source schema contains region names but no coordinates; therefore the app uses regional charts unless valid latitude and longitude are supplied, rather than inventing map points.

9. Discussion and Recommendations

The dashboard can support questions about acquisition-purpose mix by region, financing behavior by segment and common referral channels. Marketing teams should treat these observations as hypotheses and validate them using campaign response and conversion outcomes. Property analysis should be restricted to correctly linked records. The age difference is pronounced, while acquisition purpose and loan shares differ more modestly. Recommended next steps are testing cluster stability across seeds and time periods, comparing hierarchical clustering, collecting outcome labels, evaluating campaign uplift, and assessing fairness and privacy before operational use.

10. Limitations

The workflow is unsupervised and lacks ground-truth labels. K=2 is a modeling choice rather than proof of the natural number of market segments. Age may dominate the interpretation without establishing business usefulness. The schema lacks income, net worth, return on investment, portfolio size and campaign outcomes. Region labels do not provide coordinates. Missing or inconsistent client references prevent safe attribution of some property records. Results can change with data, preprocessing, reference date and initialization.

11. Conclusion

This project demonstrates an exploratory pipeline from cleaning and feature preparation to weighted K-Means clustering and interactive Streamlit analytics. The recorded two-cluster run separates buyers primarily by age and has a silhouette score of 0.3012. Acquisition purpose and loan patterns provide additional descriptive context. The dashboard makes the patterns inspectable through filters, visualizations and downloadable summaries. It should be treated as a decision-support prototype, not a validated investment recommendation system. Stronger business claims require stable cluster validation, reliable transaction-to-client linkage, geographic coordinates, richer financial attributes and outcome-based evaluation.